



BHARATI VIDYAPEETH'S COLLEGE OF ENGINEERING, NEW DELHI

Department of Computer Science and Engineering

MINI PROJECT PROGRESS REPORT

Period of Report from 15.09.2025 to 5.11.2025

Sr. No.	Student Name	Enrollment Number
01	UTKARSH CHAUDHARY	08011502723
02	YASH VERMA	08111502723
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Attendance and Feedback By Mentor

Enrollment no.	Student Name	Attendance out of 5	Feedback/Suggestion	Signature

Mentor Name: Dr. Aarti

Approved Project Topic: Agent Based Adversarial Defence Framework for Web Security

Work done during the period:

- 1. Literature Review and Domain Understanding:** We started by analyzing several high-impact papers in detail. Singh et al. 's "Hierarchical Multi-agent Reinforcement Learning for Cyber Network Defense" (2024) became our primary reference because it had exactly the red-blue adversarial agent structure we needed, plus complete implementation code on GitHub. However, it focused on network-level security (intrusion detection, DDoS), not web applications. We also studied ModSec-AdvLearn, which tackles adversarial approaches to SQL injection defense. This helped us understand web-specific challenges: how attackers obfuscate payloads, encoding tricks, and why traditional defenses struggle. Two ACM Computing Surveys on adversarial MARL and cybersecurity ML provided our theoretical foundations.

Through this review, we identified the critical gap: while hierarchical MARL works well for network defense, nobody has applied it to web application security. Most SQL injection defenses use static rules requiring manual updates and can't self-adapt to new attack techniques.

- 2. Architecture Design: Web-HMARL Framework:** We designed a three-component system called Web-HMARL (Web-focused Hierarchical Multi-Agent Reinforcement Learning):

Red Agent (Attacker): Simulates an intelligent adversary learning which SQL injection techniques bypass current defenses. Built as a Deep Q-Network generating 47 attack variants, from basic boolean injections to sophisticated obfuscated attacks. It's rewarded for discovering novel bypass techniques.

Blue Agent (Defender): A three-tier hierarchy adapted from Singh's network approach:

- Tier 1: Fast detection using lightweight neural networks
- Tier 2: Makes security vs. usability trade-offs, balancing false positives and negatives
- Tier 3: Deep asynchronous analysis using Transformers and Graph Neural Networks; findings improve Tiers 1 and 2

Master Coordination: Manages request routing, retraining triggers, and resources using expert-driven rules rather than pure learning.

- 3. Gap Validation:** Multiple searches confirmed our research gap. While papers exist using single RL agents to generate attacks or ML classifiers to detect them, none combine both into an adversarial multi-agent system for web applications. This validates that we're transferring proven network security techniques to a domain with fundamentally different constraints.
- 4. Evaluation Planning:** We established metrics capturing adversarial dynamics: detection rates for known vs. novel attacks, false positive/negative rates, convergence speed, and performance against real penetration tools like sqlmap. We'll compare ModSecurity with OWASP CRS (traditional WAF), static ML classifiers, recent deep learning approaches, and ModSec-AdvLearn. The goal is demonstrating that adversarial MARL provides superior adaptability by handling new attack patterns without manual updates.

Status / Stage of the Project:

We're currently transitioning from the design phase to implementation. We have a comprehensive blueprint with detailed specifications: state spaces, action spaces, reward functions, neural network architectures, and training procedures. Someone could theoretically take our plans and start coding immediately.

The extensive upfront work has significantly de-risked the project. We've validated that (a) the problem is worth solving, (b) similar approaches work in related domains, and (c) we have a concrete, feasible implementation plan. This positions us to execute a well-thought-out research plan rather than hoping for the best.

Next Phase: Implementation and Experimentation

We'll start by building the development environment and the vulnerable web application where agents will interact. Then comes the exciting part: implementing the agent architectures and running the adversarial training loop to see how they actually learn from each other.

Our approach is phased. We'll begin with a simplified proof-of-concept, maybe just Tier 1 defense against a basic red agent, to validate that adversarial training works in this domain. Then we'll gradually add hierarchical complexity, introduce sophisticated attack variants, and refine based on observations.

The goal is preliminary results showing that adversarial co-evolution produces more adaptive defenses than static baselines. That would be a solid contribution even if the full three-tier system isn't perfect yet. We expect new challenges to emerge during implementation, but our solid foundation means we can adapt from strength rather than constantly questioning our fundamental approach.

(Signature of Mentor)

(Signature of the Head of the Department)
